

# Hassaan Khan, EIT

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## EDUCATION

**University of Illinois Urbana - Champaign (UIUC)** 2025 - present

*Doctor of Philosophy* in Civil Engineering

- Concentration: Energy-Water-Environment Sustainability (EWES)
- **Dissertation:** Understanding the effects of hydroclimatic extremes across the electricity pipeline and figuring out what ensuring throughput during adverse operating conditions looks like as practical procedures and optimizations relevant to practitioners (*R2O - research to operations*)
- Relevant Coursework: Power System Analysis, Dynamic System Reliability, Probabilistic Risk Analysis, Data Science for the Geosciences, Climate & Social Vulnerability, Hydrocomplexity

**University of Illinois Urbana - Champaign (UIUC)** 2025

*Master of Science* in Civil Engineering

- Concentration: Energy-Water-Environment Sustainability (EWES)
- **Thesis:** *Towards connecting occupant behavior to residential water use in a permanent supportive housing context* - [publication](#)

**National University of Sciences & Technology (NUST)** 2022

*Bachelor of Engineering* in Civil Engineering

## SKILLS

**Data Analysis & Languages:** Python (pandas, scikit-learn, xarray), MATLAB, Excel (pivot tables)

**Applications & Tools:** AutoCAD, SAP2000, HEC-HMS, BASEMENT, ArcGIS Pro, EPANET, HAZUS, SWMM, ParaView, Linux Shell

## EXPERIENCE

**University of Illinois Urbana - Champaign (UIUC)** Urbana, IL

*Graduate Research Assistant (RA)* 05/2025 - present

Working on hydroclimatic extremes and infrastructure systems within the [Stillwell Research Group](#)

*Graduate Teaching Assistant (TA)* 08/2026 - 12/2026

CEE 433: Water Technology and Policy

Teaching, conducting office hours and grading assignments and exams for a section of 50 students

*Graduate Teaching Assistant (TA)* 08/2025 - 05/2026

CEE 498: Community Disaster Resilience

Curriculum development and instruction responsibilities for collaborative course on community resilience and disaster planning with stakeholders across CUPHD, ERDC, Champaign County Emergency Management Agency and campus units

*Graduate Mentor* 06/2025 - 07/2025

AREES (formerly MSI Alliance) Student Summer Research Experience

*Graduate Teaching Assistant (TA)* 01/2025 - 05/2025

CEE 350: Water Resources Engineering

Teaching, conducting office hours and grading assignments and exams for a section of 53 students

*Graduate Research Assistant (RA)* 08/2023 - 05/2025

Working on hydroinformatics and residential water within the [Stillwell Research Group](#)

## Current Water

National Science Foundation INTERN - [H2NOW Chicago](#)  
(H2NOW is a [2026 Fast Company World Changing Idea](#))

Chicago, IL  
05/2024 - 08/2024

- Created a predictive model for estimating fecal coliform levels at sites in the Chicago and Calumet rivers
- Assisted with sonde deployment and observation for 2024 season
- Prepared and cleaned previous H2NOW seasons' data for public consumption
- Developed a change-detection script to trigger alerts for turbidity spikes and possible mesh fouling to assist with O&M activities

## Educative, Inc.

Editorial (Technical Content Reviewer)

Bellevue, WA  
10/2022 - 04/2023

- Edited and reviewed technical course content across 10+ technology areas
- Developed interactive scripts to support technical courses
- Provided input for content curation and development

## National University of Sciences & Technology (NUST)

Teaching Assistant (TA)

Islamabad, PK  
09/2021 - 01/2022

- [CE-308 Plain and Reinforced Concrete](#)  
Assisted with marking quizzes, assignments and assessments; responsible for 70 students
- Helped refine assessment parameters which led to improved test scores for 10+ students

## Pakistan Council of Research In Water Resources (PCRWR), MoST

Undergraduate Intern

Islamabad, PK  
07/2021 - 09/2021

- Assisted hydrology team with supply, demand and water stress analysis for sector I-10 in consultation with the CDA
- Supported fieldwork with geotagging, classifying and quantifying process parameters for groundwater resources and distribution networks
- Analysed and interpreted results of 7+ designs and scenarios using Hydrus; facilitated R&D operations to explore the efficacy of managed aquifer recharge (MAR) systems in urban settings
- Assessed system reliability and suggested improvements to reduce system downtime by 15% and improve resilience

## Spectra Magazine

Editorial Board | [Co-Lead Interview Team](#)

08/2019 - 09/2022

- Led and steered a skilled team of 15+ interviewers and science writers
- Curated popular science content and managed proposals across four teams
- Collaborated with a team of five editors to uphold editorial standards

## CERTIFICATIONS

[Adult and Pediatric First Aid/CPR/AED](#)

[FE Exam](#) (TBPELS EIT# 86524)

PE Exam Passed (WRE - July, 2026)

## AWARDS

UIUC Academy for Excellence in Engineering Education (AE3) Teaching Award - Spring 2026

## VOLUNTEERING & SERVICE

Engineering Open House

[Special Exhibits Volunteer](#)

04/2024 - present

H2NOW Technical Advisory Committee

Member

05/2024 - present